

Malware proposal 1

Title

A Hybrid CNN–Vision Transformer Approach for IoT Malware Image Classification

1. Introduction

Malware detection is a critical problem in cybersecurity, traditionally addressed using signature-based and classical machine learning techniques. With the growth of deep learning, image-based malware classification—where binary files are converted into images—has shown strong performance, particularly using Convolutional Neural Networks (CNNs).

Most existing research in this area focuses on **Windows-based malware datasets**, such as the Maling dataset. However, modern cyber threats are increasingly shifting toward **IoT (Internet of Things) devices**, which predominantly run **Linux-based ELF binaries** and exhibit significantly different structural characteristics.

This project aims to explore whether modern deep learning architectures can effectively generalize from traditional Windows malware to **modern IoT/Linux malware**.

2. Problem Statement

CNN-based models are effective at extracting local texture features from malware images. However, due to their **local receptive field**, CNNs struggle to capture long-range dependencies within binary files.

Recent research has shown that **Vision Transformers (ViTs)** and **hybrid CNN–Transformer architectures** can model global dependencies using self-attention mechanisms. While these hybrid models demonstrate strong performance on Windows malware datasets, their effectiveness on **IoT/Linux malware remains underexplored**.

3. Research Gap

- Most malware image classification research uses **outdated Windows malware datasets** (e.g., Malimg).
 - IoT malware uses **Linux ELF binaries**, which differ significantly from Windows PE binaries.
 - Existing **Hybrid CNN–Vision Transformer models are rarely evaluated on IoT malware datasets**.
 - There is limited experimental analysis on whether such hybrid models **generalize across operating systems and malware formats**.
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4. Objectives

1. To implement a **Hybrid CNN–Vision Transformer (CNN–ViT) architecture** for malware image classification.
 2. To evaluate the model on a **modern IoT/Linux malware image dataset** (such as MaleVis).
 3. To compare the hybrid model with a **CNN-only baseline**.
 4. To analyze the impact of global context modeling on IoT malware detection.
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5. Supporting Literature (Related Work)

The proposed work is motivated by the following key studies:

1. **Nataraj et al. (2011)** demonstrated that malware binaries can be effectively represented as grayscale images and classified using visual patterns. This work introduced the **Malimg dataset**, which became a benchmark for malware image classification.
2. **Dosovitskiy et al. (2021)** introduced the **Vision Transformer (ViT)** architecture, showing that self-attention mechanisms can model long-range dependencies in visual data more effectively than CNNs alone.
3. **Hybrid CNN–Transformer approaches** have been proposed in recent studies to overcome CNN limitations by combining local feature extraction with global context modeling. These models have shown improved robustness against obfuscated and polymorphic malware when evaluated on Windows malware datasets.

4. **MaleVis Dataset (2022)** introduced a large-scale **IoT/Linux malware image dataset**, containing ELF-based malware samples converted into images, addressing the lack of modern datasets for IoT malware research.
5. Recent IoT malware studies using CNN-based models have achieved high accuracy, indicating that image-based techniques are applicable to IoT malware; however, **Transformer-based and hybrid approaches remain insufficiently explored in this domain.**

These studies collectively motivate the need to evaluate **Hybrid CNN–Vision Transformer architectures on IoT malware datasets**, forming the basis of this project.

6. Proposed Methodology

1. Dataset

- Use a publicly available IoT malware image dataset (e.g., MaleVis).
- Images represent Linux ELF malware binaries converted into visual form.

2. Model Architecture

- CNN block for local feature extraction.
- Feature reshaping (tokenization) to form a sequence.
- Vision Transformer encoder with self-attention for global dependency modeling.
- Fully connected layer for classification.

3. Evaluation

- Metrics: Accuracy, Precision, Recall, F1-score.
 - Comparative analysis with a CNN-only baseline.
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7. Expected Outcomes

- Validation of Hybrid CNN–ViT effectiveness on IoT malware images.
- Comparative insights between CNN-only and hybrid models.
- Experimental evidence on cross-platform generalization of malware detection models.

8. Significance of the Work

This project contributes by **experimentally validating existing deep learning architectures in a modern and relevant threat environment**. The findings will help assess whether state-of-the-art malware detection models remain effective as malware shifts from desktop systems to IoT ecosystems.

9. Conclusion

By applying a Hybrid CNN–Vision Transformer model to IoT malware image classification, this work addresses a relevant research gap. It provides empirical insight into the generalization capability of modern deep learning models across different malware platforms.

10. References (Sample)

1. Nataraj, L., et al., *Malware Images: Visualization and Automatic Classification*, VizSec, 2011.
 2. Dosovitskiy, A., et al., *An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale*, ICLR, 2021.
 3. Atitallah, S. B., et al., *MaleVis: Malware Evaluation with Vision Dataset*, 2022.
 4. Recent studies on Hybrid CNN–Transformer models for malware classification (2023–2025).
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